International Journal of Environmental Science and Technology
https://doi.org/10.1007/s13762-023-05016-2

ORIGINAL PAPER

Machine learning-based prediction of air quality index and air quality grade: a comparative analysis

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Received: 7 November 2022 / Revised: 19 April 2023 / Accepted: 20 May 2023
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Abstract

The purpose of this study was to compare different machine learning models for predicting daily air quality index (AQI) and evaluating air quality grade (AQG). The study used publicly available data from 2014 to 2019 for six pollutants (PM_{10}, $PM_{2.5}$, NO_2, SO_2, CO, O_3). Four models (random forest (RF), gradient boosting (GB), Lasso Regression (LASSO), and the Stacked Regressor) were used for predicting AQI, while six models (K-Nearest Neighbors (KNN), support vector machines (SVM), decision tree (DT), multilayer perceptron (MLP), random forest (RF), and the Stacked Classifier) were used for forecasting AQG. The individual models were evaluated using different statistical measures, such as *R*-squared (R^2), root mean square error (RMSE), mean absolute error (MAE), accuracy score (ACC), Matthew's Correlation Coefficient (MCC), and F1 score. The study found that the stack model performed consistently across all metric scores for AQI prediction. The stack model had an R^2 score of 0.973, RMSE of 7.568, and MAE of 4.596, outperforming LASSO, GB, and RF. This indicates that the stack model was able to minimize the weaknesses of the individual models and provide a more accurate prediction. For AQG, the stack model also performed better across all metric scores, with an ACC of 0.970, MCC of 0.960, and F1 of 0.970, outperforming MLP, KNN, SVM, DT, and RF. The study concluded that stacked generalization machine learning models can be used for forecasting air quality index and grade with high efficiency and precision, mitigating the concerns of overfitting against individual models.

Keywords Air quality prediction · Stack models · Forecasting · Air quality classification · Machine learning

Introduction

Air is one of the main resources for the sustenance of life (Akinfolarin et al. 2017; Mohammed and Caleb 2014). Air pollution is a major environmental issue globally, and it affects the quality of the atmosphere and the health of humans (Kurt and Oktay 2010; Mahalingam et al. 2019). Rapid industrialization and urbanization have led to increased levels of air pollution, and the emission of industrial pollutants is a major cause of severe air pollution in industrialized areas (Zhu et al. 2017). The pollution of air

Editorial responsibility: Chenxi Li.

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Published online: 07 June 2023

and its related problems have resulted in increased public attention to air quality in both developing and developed countries. The quality of the atmospheric environment is increasingly attracting public concern because the quality of air affects everyone in society. This has resulted in the need for strategies in air quality management and pollution control.

Many air pollutants such as carbon monoxide (CO), sulfur dioxide (SO_2), nitrogen dioxide (NO_2), suspended particulate matter (SPM), and ozone (O_3) adversely affect the quality of the atmosphere and negatively impact human health (Sowlat et al. 2011). High concentrations of these atmospheric pollutants can be life-threatening, causing many health problems including breathing difficulty, headache and dizziness, asthma, impaired lung function, and cardiopulmonary illnesses (Banerjee and Srivastava 2011; Kumar and Goyal 2011a; Zhu et al. 2017). Clean air can be polluted by many anthropogenic sources including domestic smoke and heating or combustion devices, motor vehicles and aircraft pollution, incineration of waste, and outdoor fires such as forest fires (Mishra and Goyal 2015; Soni and Patel 2018). These activities release a good amount of gaseous emissions (SO_2, NO_2, CO, H_2S, VOCs, and hydrocarbons) and particulates (smoke, soot, metallic, dust, fumes, and aerosols) which affects air quality. The combustion of motor vehicle fuels (fossil fuels) contributes to most of these pollutants (Banerjee and Srivastava 2011; Soni and Patel 2018).

In China, atmospheric pollution is a significant concern, and it has led to the enactment of the National Ambient Air Quality Standards and the implementation of the Air Pollution Prevention and Control Action Plan to cut down $PM_{2.5}$ concentration, which has been reported to be the primary air pollutant in major cities in China (Chen et al. 2015; Qiao et al. 2015). Concerning the measurement of air pollution, a total air pollution measure that accounts for all major pollutants is needed to assess the overall air quality and the general trends in the ambient air quality in different localities. In this regard, the Air Quality Index (AQI) is a comprehensive tool that serves as an indicator of air pollution levels based on multiple air pollutants. Compared to single pollutant indicators, AQI makes it easier for the public to understand how bad or good the quality of air is and to promote their awareness of air pollution management.

AQI is essential in drawing attention to the daily and monthly variations in ambient pollutant concentrations and so contributes to education and awareness of air quality issues, and communicates forewarning information on environmental trends and potentially health-damaging air pollution events to the public and policy-makers (Kumar and Goyal 2011b). It does not only indicate the period of danger but also informs on what contingency plans are to be implemented on occasions of polluted index rating. AQI also helps to forecast and give advanced information about changes in the concentration level of air pollutants (Mishra and Goyal 2015). Air quality indexes are widely used in air quality management schemes which help to classify air quality grades (AQG) according to the concentrations of pollutants in the air and report it with expressions like "good", "moderate pollution", or "severe pollution".

Most previous studies on air pollution mainly focused on a single pollutant such as particulate matter and SO_2. However, people are seldom exposed to a single pollutant (Choi et al. 2016; Dominici et al. 2010). To assess the overall air quality and the general trends in the ambient air quality in different localities, a total air pollution measure that accounts for all major pollutants is needed. In this regard, a simple and effective tool, such as an air quality index, is needed to provide timely information about the status of air quality for the assessment and compliance with reference standards and to also evaluate the effectiveness of emission control policies. The AQI is a comprehensive tool that serves as an indicator of air pollution levels based on multiple air pollutants (Bao et al. 2015). Compared to single pollutant indicators, AQI makes it easier to understand how bad or good the quality of air is and to promote awareness of air pollution management (Kumar and Goyal 2011a; Pan et al. 2020). This can be regarded as an index or rating scale used to evaluate and report the air quality status through a quantitative measure by aggregating and summarizing air quality information based on available data to illustrate the existence and trend in concentrations of many air pollutants into a single numerical form (Pan et al. 2020; Zhu et al. 2017).

Research into models for predicting AQI is gradually increasing with several statistical and machine learning models being developed to forecast air quality. In recent years, a number of machine learning-based air pollution prediction models have been used to estimate varying levels of air pollution at various places. Zhu et al. (2019) proposed a two-step head forecasting model based on complementary ensembles empirical mode decomposition (CEEMD), grey wolf optimizer (GWO), and support vector regression (SVR) to predict the two major sources of acid rain (NO_2 and SO_2) in four cities of Central China. Cheng et al. (2019) also used a Wavelet-ANN model to forecast short-term $PM_{2.5}$ concentrations in China. Different regression models such as Support vector regression (SVR) and multiple linear regression model were also used by Ganesh et al. (2017) to predict AQI by using the concentrations of NO_2, CO, O_3, $PM_{2.5}$, $PM_{10,}$ and SO_2 in New Dhali. Ganesh et al. (2017) indicated in their study that, SVR exhibited high performance in terms of the investigated measures of quality. This was also supported by Janarthanan et al. (2021) who combined

SVR with long short-term memory (LSTM) to classify air quality index values based on deep learning models. SVR together with random forest regression (RFR) model was further employed by Liu et al. (2019) to predict Nitrogen oxides (NO_X) concentration in an Italian city and the AQI in Beijing. Their study showed that the SVR-based model has a better performance in the prediction of the AQI, while the RFR-based model performed better in predicting NO_X concentrations. A novel hybrid model was also used by Liu and Chen (2020) to predict the spatial variation of AQI while assessing the spatial correlation between monitoring stations, the relationship among air pollutants, and the lagged effects of historical time series.

The accurate predictions of AQIs are of enormous importance to governments, enterprises, and the general public in making informed decisions. Although several models have been proposed for air quality prediction, the issue of accuracy in their prediction has not been fully addressed among the existing models, thus bringing the efficiency of proposed models to question. For instance, the principle of statistical models such as multiple linear regression and autoregressive integrated moving average model may be simple and efficient (Dastoorpoor et al. 2018), however, their prediction accuracy is still low (Taylan 2017; Wu and Lin 2019). Similarly, artificial intelligent models alone such as least squares support vector machines, artificial neural networks, or even support vector machines, are unable to provide satisfactory and accurate predictions due to the non-stationary nature of the concentrations of the pollutants (Zhu et al. 2017). The proposed hybrid models try to combine different models to perform air quality predictions, however, there is still a lack of research into the effectiveness and accuracy of the hybrid models and as such its rationality still needs further verification (Wu and Lin 2019). This has resulted in the unavailability of a universally accepted or standardized model for the prediction of air quality. There is therefore a need to develop an effective system or technique that will address the weakness of individual models and overcome the existing problems of prediction accuracy associated with air quality prediction.

This study utilizes individual models, such as K-nearest neighbors (KNN), support vector machines (SVM), decision tree (DT), random forest (RF), multilayer perceptron (MLP), gradient boosting (GB), and LASSO to make predictions. It further ensembles these models in base learners to forecast AQI and AQG for the next day (single-step forecasting) by running the output of multiple models through a "meta-learner" (linear regressor and classifier). The results for the single model predictions are then compared with the ensemble model. This helps provide robust and valuable information on the machine learning models suitable for providing timely information, with minimal errors, on the trend of ambient air quality to the public so that precautionary measures to protect lives can be taken.

Materials and methods

Data source and pre-processing

The entire dataset used in this study was obtained from the China National Environmental Monitoring Centre (CNEMC, http://www.cnemc.cn). The dataset is made up of A 6-year daily air pollutants data (2014–2019) for Beijing. Six air pollutants namely Particulate Matter (PM_{10}, $PM_{2.5}$), Nitrogen dioxide (NO_2), Sulphur dioxide (SO_2), Carbon monoxide (CO), Ozone (O_3), and their corresponding AQI and AQG were used in this study. These pollutants are known to have adverse effects on health and the environment (Abelsohn et al. 2002), and have been identified to be useful communication parameters that can be used to translate air pollution information that the public can understand (Sonibare et al. 2010).

Initial screening of the data revealed that there were no missing values in the dataset. When using the interquartile range method (IQR) for detecting outliers, we found few observations (6) to be outliers. These observations were removed from the dataset since the outliers were not many to influence the study. From Table 1, strong positive correlations were observed between $PM_{2.5}$ and PM_{10}, $PM_{2.5}$ and CO, and $PM_{2.5}$ and NO_2. On the other hand, moderately strong correlations were observed among $PM_{2.5}$ and SO_2, PM_{10} and NO_2, SO_2 and CO, and CO and NO_2.

Table 1 Correlation matrix for pollutants

	$PM_{2.5}$	PM_{10}	SO_2	CO	NO_2	O_3
$PM_{2.5}$	1.000	0.847	0.550	0.844	0.782	−0.094
PM_{10}	0.847	1.000	0.533	0.714	0.730	−0.048
$S0_2$	0.550	0.533	1.000	0.626	0.615	−0.272
CO	0.844	0.714	0.626	1.000	0.804	−0.297
NO_2	0.782	0.730	0.615	0.804	1.000	−0.339
O_3	−0.094	−0.048	−0.272	−0.297	−0.034	1.000

Feature selection

The first step of automating the estimation of AQI and AQG is to select the important pollutant variables. The idea of feature selection in machine learning is to remove irrelevant variables to improve the efficiency of the models (Khosravi et al. 2018; Bui et al. 2019). For this study, feature selection by variance threshold and mutual information was employed for the regression and classification models. Feature selection outcomes from the variance threshold and feature selection by mutual information (see Fig. 1) indicated that PM_{10}, O_3, CO, NO_2, and SO_2 were the important features in estimating AQI and AQG. $PM_{2.5}$ was the most important, followed by PM_{10}, O_3, CO, NO_2, with SO_2 having the least significance.

Preparation of datasets for the single-step (one day) prediction scenario

This current study used AQI and AQG as the output. $PM_{2.5}$, PM_{10}, SO_2, CO, NO_2, and O_3 were used as the inputs. Note that the AQI variable data was a continuous type for the regression analysis. The classification of the AQG was also undertaken in this study. To automatically forecast the AQG after predicting the AQI, another dataset was prepared. Here, the output was transformed from the continuous air quality index value (regression datasets) into classes "excellent", 'good", "light pollution", "moderate pollution", "serious pollution" and "severe pollution" for the classification datasets. In this study, the outputs values for the air quality grade were labeled as 0–50 "excellent", 51–100 "good", 101–150 "light pollution", 151–200 "moderate pollution", 201–300 "serious pollution" and 301–500 "severe pollution". These were chosen based on the Chinese national classification system. A one-day scenario was designed to evaluate the air quality index and grade. In other words, the current or today's concentrations of $PM_{2.5}$, PM_{10}, SO_2, CO, NO_2, and O_3, were used to predict AQI and AQG for the next day (tomorrow). Figure 2 shows the preparation of the datasets.

Model optimization

Model optimization is one of the most important processes for better results with machine learning models. Optimization finds values of the variables that minimize or maximize the objective function while satisfying the constraints, that is, minimizing error, cost, or loss when fitting the model. This involves selecting a set of optimal hyperparameters for the learning algorithm and the performance of the machine learning models is directly related to the hyperparameters, which are set before training. In this study, some hyperparameter optimization was included to make the models perform better on the validation data set. The stack models used for this study automatically tune the hyperparameters for the different sub-models through continuous iterative calculations by grid search Cross Validation (CV) and random parameter CV. The number of layers and hidden neurons, regularization parameter (c), kernel coefficient (gamma), kernel type, max_depth,

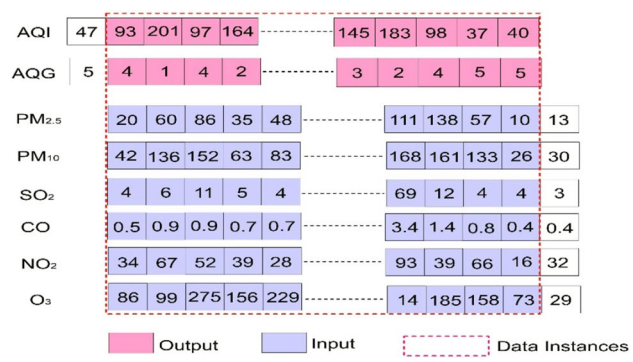

Fig. 2 Data scenario for one day (single step) forecasting

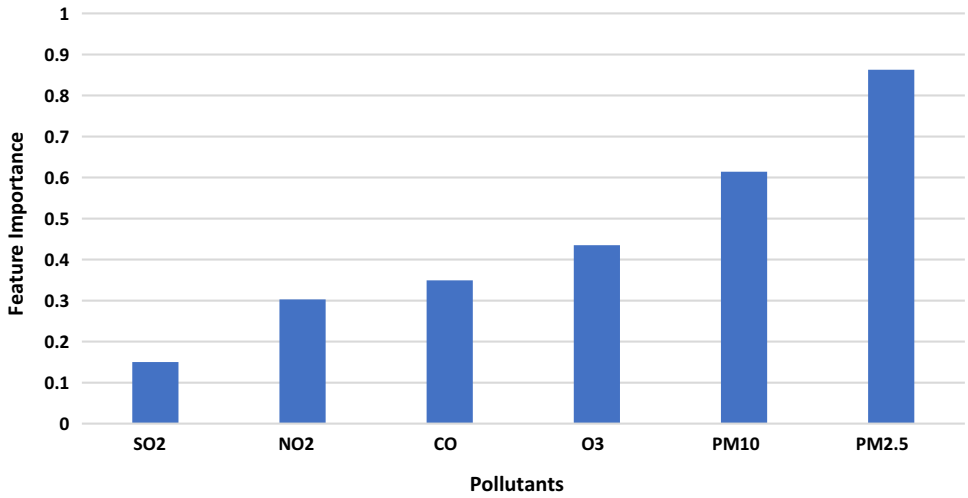

Fig. 1 Feature importance for forecasting AQI and AQG

and n_estimators are some of the hyperparameters used in this study.

Training and testing dataset

The entire dataset was split into training and testing sets. The testing set, also known as the independent testing set, was used to verify the generalization capability of the trained models. The preparation of training and testing sets for the regressing was split in an 80% and 20% ratio: 80% for the training and the remaining 20% for testing. However, the preparation of training and testing sets for the classification was performed using random splitting.

Operationalised models

K-nearest neighbors

K-nearest neighbors is a supervised machine learning algorithm that can be used to solve both classification and regression problems. In regression, the output is a real number while in classification it gives a discrete value as its output. In this method, it is assumed that similar items exist in propinquity and are therefore considered to be related. The method uses distance measures such as the Euclidean distance, Manhattan distance, etc. to estimate how close the items or objects are related to each other. The distance measure used has a great impact on using the KNN classifier (Alfeilat 2019). The Euclidean distance was used for this research and can be defined as the distance between two points in a straight line in Euclidean space, mathematically expressed as:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} \left(x_i - y_i\right)^2} \tag{1}$$

In this study, the k was determined by using a grid search to get the best k value that could help classify the new labels into the appropriate classes.

Support vector machines

Support vector machines are supervised machine learning models that can be used for classification and regression analysis, it is used in classification to classify objects. The classes are separated by a line called a separating hyperplane, and the points that are closest to the separating hyperplane are termed support vectors. The hyperplane is of the form $w^T x + b$, to obtain the label for the classification problem, the input data has to be passed to a function such as sigmoid, Heaviside step function, etc. and the output will be the label that is used in the optimization process. To find the value of w and b, the support vectors must be identified by finding the smallest margin by maximizing the margin (Harrington 2012):

$$\arg\max_{w,b} \left\{ \min_{n}(labe \lg(w^T x + b))g \frac{1}{\|w\|} \right\} \tag{2}$$

For the support vector machine to be used for a large dataset the kernel has to be defined. During the optimization of the parameters, three kernels were defined but the radial bias function emerged as the best kernel for this classification problem. The radial bias function using the Gaussian approach can be defined as (Harrington 2012):

$$k(x, y) = \exp\left(\frac{-|x - y|^2}{2\sigma^2}\right) \tag{3}$$

where σ defines how fast it gets to 0.

The values of the parameters for this study were obtained using a grid search to get the best parameters C and γ using cross-validation.

Decision tree

Decision Tree is a supervised machine learning technique that is based on rules in making decisions (Kamiński et al. 2018). In Decision Tree, the features of the dataset are used to create yes or no questions and continually split the dataset till items fall into classes. The structure of the data is termed a tree structure and anytime a yes or no question is passed a node is created. The first node created is called the root node and the last node is called the leaf node. Whenever a question is answered, it creates branches and segments the feature space into disjoint regions. This can be seen in Fig. 3.

Random forest

Random Forest is a supervised machine learning algorithm used for both classification and regression analysis (Ho 1995). It makes use of decision tree classification since it involves a collection of decision tree classifiers before the final decision is made. The collection of the classifiers is termed the forest. Random Forest unlike other algorithms does not require tuning of hyper-parameters and is free from overfitting since it considers a comprehensive number of Decision Trees before making the final decision. Compared to Decision Tree, Random Forest is difficult to interpret and very slow in prediction, and computationally time-consuming. During the splitting process in Random Forest, the algorithm searches for the best feature in a subset of random

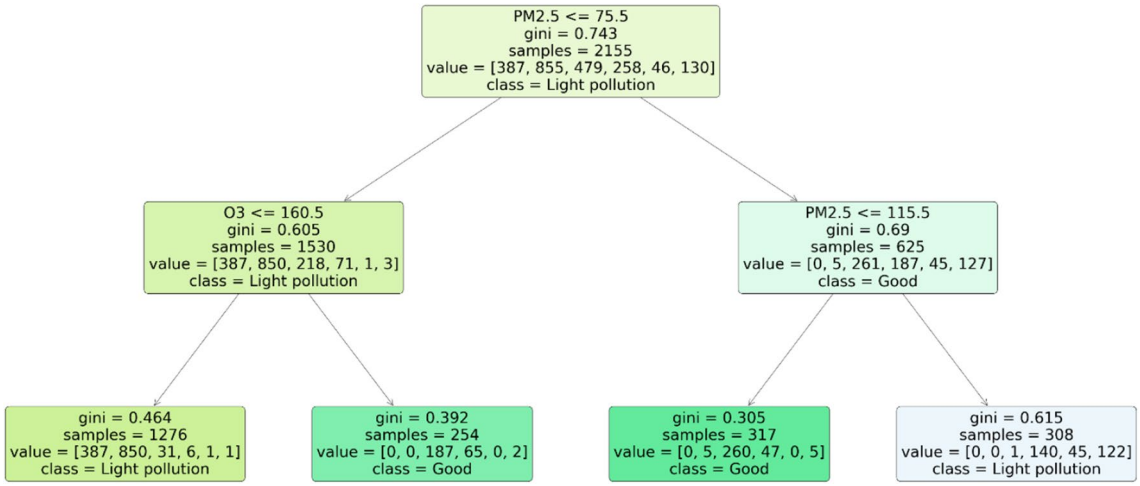

Fig. 3 Example of decision tree illustrating how the features are segmented until a leaf note is obtained

features rather than searching for an important feature while splitting a node. This results in obtaining a better model.

Multilayer perceptron

Multilayer Perceptron (MLP) is a feedforward neural network that comprises three layers: the input layer, the hidden layer, and the output layer (see Fig. 4). In a multilayer perceptron, the neuron is combined weights that are subject to an activation function such as sigmoid, ReLU, etc. The activation function has to be differentiable to use gradient descent. The algorithm uses backpropagation intending to minimize cost.

The feedforward algorithm in MLP is obtained by calculating the value of the hidden unit which can be expressed as:

$$z = \sum_{i=1}^{n} w_{ij} x_i, \quad j = 1, 2, \ldots, n \tag{4}$$

The activation function is then applied to the z value, given the sigmoid activation which is expressed as: $\phi(z) = \frac{1}{1+e^{-z}}$, the activation unit is expressed by the equation without the bias term as:

$$a = \phi\left(\sum_{i=1}^{n} w_{ij} x_i\right) \tag{5}$$

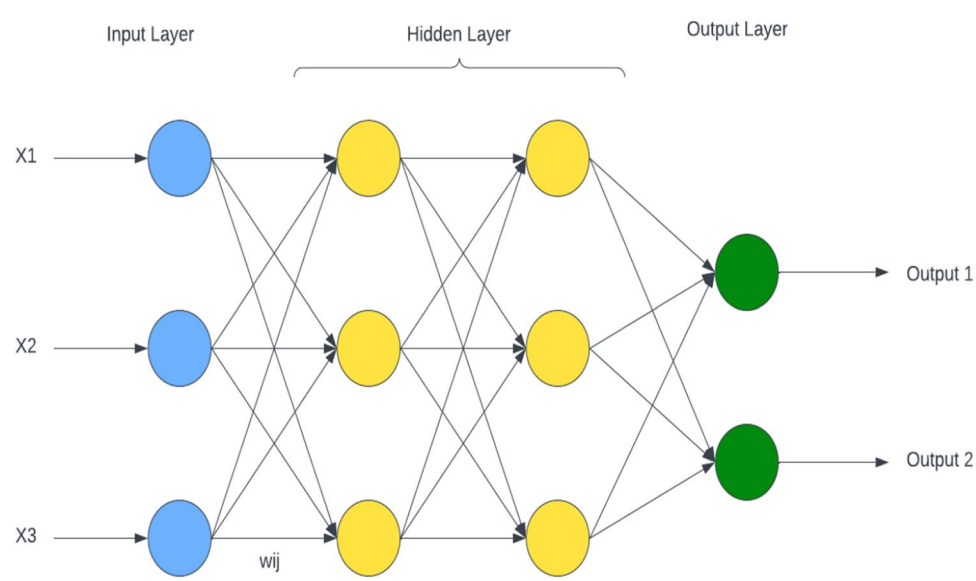

Fig. 4 Feedforward neural network architecture for MLP

Gradient boosting

Gradient Boosting is a machine learning algorithm that can be used for both classification and regression analysis. It is formed as a collection of the ensemble of weak models, the main aim is to minimize loss by combining weak learners. When the target is continuous it is termed Gradient Boosting Regressor and when applied in classification it is termed Gradient Boosting Classifier.

Lasso regression

Lasso regression is a regression technique that uses the shrinkage method. It is applied to aid in increased model prediction. Lasso automatically performs feature selection, as a result, it is mostly used when there are a lot of features as it reduces the number of features. In Lasso regression, a penalty term which is the absolute of the coefficient term is added to the regression model to regulate the bias and the error; this is termed regularization. The Lasso regression regularizer is referred to as the L1 norm. The lasso regression can be expressed in a mathematical term:

$$\sum_{i=1}^{n}\left(y_i-\sum_{j}x_{ij}\beta_j\right)^2+\lambda\sum_{j=1}^{p}\left|\beta_j\right| \tag{6}$$

$$\sum_{i=1}^{n}\left(y_i-\hat{y}\right)^2+\lambda\sum_{j=1}^{p}\left|\beta_j\right| \tag{7}$$

where λ is the amount of shrinkage.

Stacked models

The Stack Model is a collection of different base models with the objective of increasing the performance of the model. The stack model can be used for both classification and regression analysis. In a classification analysis, the stack model is termed a classifier while in a regression analysis it is termed a regressor. Stacked models also known as Stacked generalization were first introduced by Wolpert (1992) and have also been made popular in current times by Breiman (1996) and Polley et al. (2019). In stacking, the individual base models are trained separately then afterward these models are stacked together. The study trained the individual models namely: k-nearest neighbors, Support vector machines, Decision Tree, Random Forest, Multilayer Perceptron, Gradient boosting, and LASSO. After training, these models were ensembled to obtain the stacked model for this study. Figure 5 demonstrates the operationalization of the framework for the automated AQI and AQG forecasting system.

Evaluation of the system performance

The development of models for predicting accurate AQI is progressing in the research community. Several statistical and machine learning models have been used (see Mahalingam et al. 2019; Feng et al. 2013; Wang et al. 2016; Dragomir 2010) however, no universal method has been accepted or standardized. It is, however, prudent to experiment with new and or innovative methods and techniques for better predictions and performance. The main objective of this study was to experiment with different machine learning models to forecast AQI and AQG and further stack these models in base learners to determine the best-performing model for forecasting AQI and AQG. In this study, all the individual models, namely KNN, SVM, DT, MLP, GB, and LASSO, were applied independently and ensembled with a stacking regressor (linear regression) and a stacking classifier (logistic regression) to forecast a one-day AQI and it associated AQG. For the AQI estimation, 3 individual models, namely GB, DT, and LASSO were operationalized and evaluated. These models were then assembled in a stacking regressor (linear regression) and further assessed. The R-squared (R^2) statistical measure, the Root Mean Square Error (RMSE), and the Mean Absolute Error (MAE) scores were used for the evaluation of the linear models. R^2, RMSE, and MAE methods are known to be the best methods to assess the precision, efficiency, and accuracy of linear models (Ben Seghier et al. 2020; Zhang et al. 2013; Ahmadi et al. 2020). For the AQG classification, individual models, namely KNN, SVM, DT, and MLP were applied and ensembled with a logistic regression stacking classifier. Accuracy score (ACC), Matthew's Correlation Coefficient (MCC), and F1 score were used for validation of the multiclass classification. ACC, MCC, and F1 methods are known to be the best methods to assess the efficiency and accuracy of multiclass classification (Chicco and Jurman 2020; Canbek et al. 2021). To comparatively evaluate the robustness and predictive power of the individual models and the ensemble model, the performance of the classification models for the training and testing datasets was assessed. It is important to assess the outcome of the training and testing datasets to comparatively evaluate the efficiency and accuracy of the individual models and the stack model. These training results denote the robustness of the models while the testing outcome depicts the predictive power of the models (Khosravi et al. 2018).

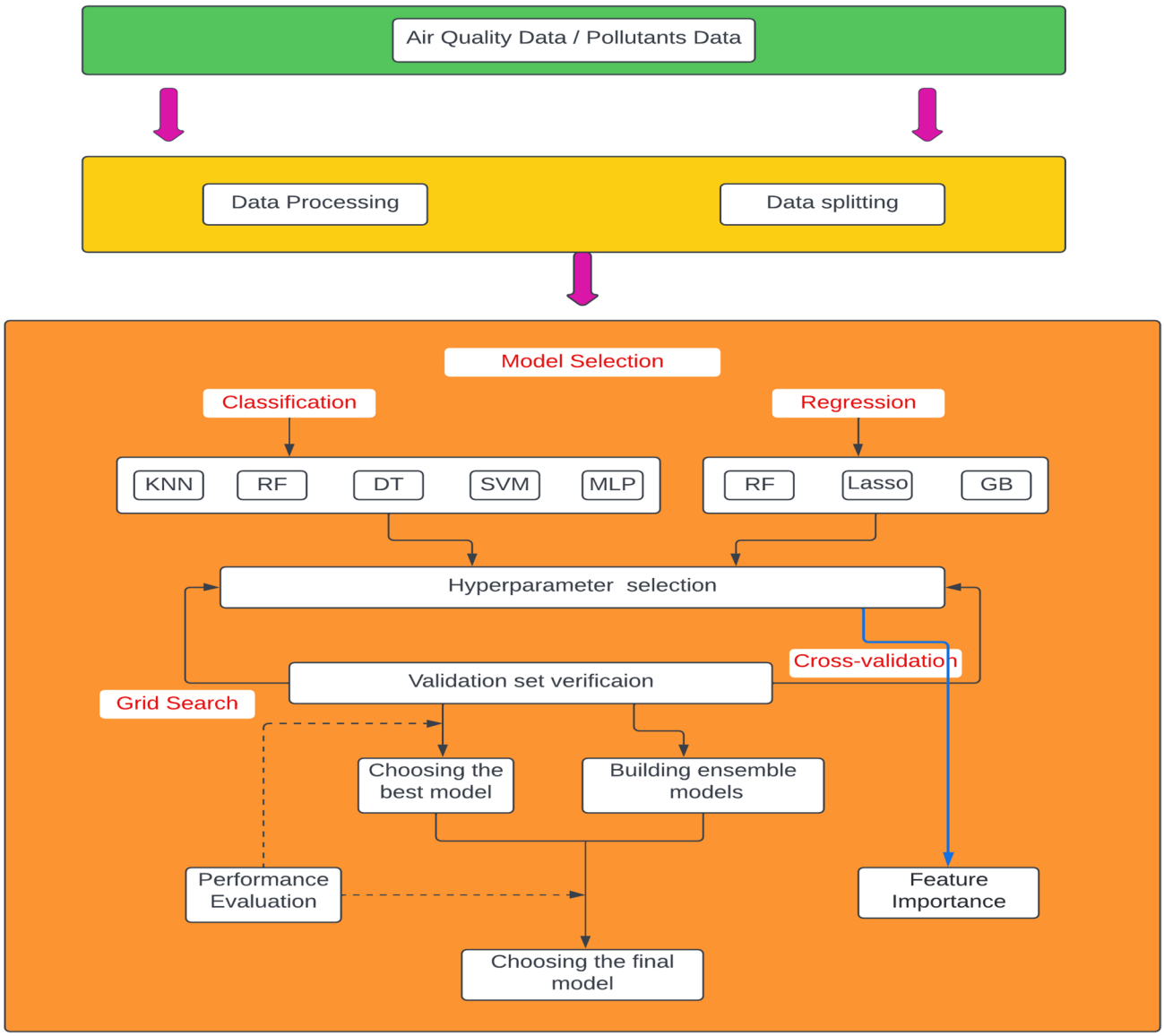

Fig. 5 Framework for the automated AQI and AQG forecasting system

Coefficient of determination

The coefficient of determination (R^2) is the measure of the contribution of the independent variable(s) to the dependent variable. It explains the percentage of variation in the dependent variable predictable by the independent variable. The coefficient of determination measures how well the dependent variable is replicated by the model. It normally ranges from 0 to 1. It can be represented mathematically in terms of the correlation coefficient as:

$$R^2 = 1 - \left(\frac{\sum_{i=1}^{n} \left(\hat{y} - y_i \right)^2}{\sum_{i=1}^{n} \left(\bar{y} - y_i \right)^2} \right) \tag{8}$$

Root mean square error

The root mean square error is the square root of the mean square error, it has the same unit of the quantity plotted on the vertical axis. The mean square error measures how a set of points deviate from the regression line, the smaller the error the closer the points are to the fitted regression line. It is an average of the square of the difference between the observed and the predicted values. It can also be calculated by finding the average of the sum of squares error. Mathematically, the Mean Square Error (MSE) can be expressed as:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2 \tag{9}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{10}$$

Mean absolute error

The mean absolute error is the average of the absolute difference between the observed values and the predicted values. It can be expressed mathematically as:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y - \hat{y}\right| \tag{11}$$

Matthew's correlation coefficient (MCC)

The MCC is a statistical tool used for model evaluation that measures the difference between the predicted values and the actual values. Its equivalent statistical measure is the chi-squared statistics for a 2 × 2 contingency table. It is the best single-value classification metric that summarizes the confusion matrix. The MCC like other correlation coefficients lies between + 1 and −1. The formula for the MCC based on the confusion matrix groups (true positives (TP), false negatives (FN), true negatives (TN), and false positives (FP)) is given as:

$$\mathrm{MCC} = \frac{\mathrm{TN} \times \mathrm{TP} - \mathrm{FN} \times \mathrm{FP}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}} \tag{12}$$

Research by Chicco and Jurman (2020) concluded that MCC is the best accuracy measure for binary classification compared to the F1 score and Accuracy measure.

F1 score and accuracy

F1 score is an accuracy measure that combines the precision and recall of a test or classifier into a single metric and finds their harmonic mean. The F1 score can be expressed as:

Precision $= \frac{\mathrm{TP}}{\mathrm{TP+FP}}$ and Recall $= \frac{\mathrm{TP}}{\mathrm{TP+FN}}$

$$\text{F1 score} = \frac{2(\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \tag{13}$$

Accuracy is a measure of the number of items that have been correctly classified. The formula for accuracy in classification is given as:

$$\text{Accuracy} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FP}+\mathrm{TN}+\mathrm{FN}} \tag{14}$$

Results and discussion

A 6-year air pollutants data (2014–2019) of Beijing are employed to evaluate the effectiveness of the proposed automated AQI and AQG estimation and forecasting system. The fundamental concept of this automated system is to forecast 24 h AQI and AQG using the current air pollutants data. The summary statistics for the pollutants used in this study are shown in Table 2. Particulate and gaseous pollutants that are of major public health concern continue to pose a serious threat to human health when their concentration exceeds the permissible limits prescribed by the regulatory authorities. The maximum values of all pollutants observed in this study exceeded the current ambient air quality standards in China,

Table 2 Summary statistics of air pollutants

	$PM_{2.5}$	PM_{10}	SO_2	CO	NO_2	O_3
Count	2155	2155	2155	2155	2155	2155
Mean	64.34385	90.25197	10.2464	1.038794	45.50998	98.26357
Std.	59.65299	67.48463	13.67935	0.842801	22.59886	63.19687
Min	0	0	0	0	0	0
25%	24	44	3	0.6	30	52
50%	47	75	5	0.8	40	83
75%	84	117	11	1.2	56	141
Max	477	550	133	8	155	311

as specified in GB 3095–2012 (2012). The health effects of air pollutants largely depend on the type of pollutant, its concentration in the environment, and the duration of exposure to the pollutant (Sicard et al. 2011). These concentrations depend not only on the quantities that are emitted from their polluting sources but also on the ability of the atmosphere to either absorb or disperse such emissions (Soni and Patel 2018). Different pollutants can also have different exposure–response characteristics; O_3 can enter the body through inhalation which can reach the respiratory system which can consequently induce changes in lung function, airway inflammation, and increased airway responsiveness to broncho-constrictors (Sicard et al. 2011). SO_2 in a gaseous form can irritate the respiratory system, effect on lung functioning may occur in the case of short-term high exposure. NO_2 exposure at high concentrations can also lead to rapid and severe lung damage (Maynard 2004). Particulate pollutant is known to have adverse health effects on plants, animals, humans, and materials in the form of visibility reduction and soiling of buildings (Soni and Patel 2018). $PM_{2.5}$ is known to be associated with respiratory problems and infections including asthma, lung cancer, and other cardiovascular and respiratory diseases (World Health Organization 2013). Another negative effect of these pollutants, especially SO_2 is the formation of acid rain, which consequently affects plants, soils, rivers, and wildlife (Mishra and Goyal 2015).

Comparison of AQI models

The results from the training datasets, as presented in Fig. 6, showed that RF achieved the highest score for the proportion of variance in AQI explained by the independent variables ($R^2 = 0.991$). Additionally, RF had the least prediction error (RMSE = 6.590) and the least magnitude of difference between the predicted and true AQI values (MAE = 1.048). On the other hand, LASSO had the lowest scores across all performance evaluation metrics. The individual models can be ranked by their order of magnitude of accuracy

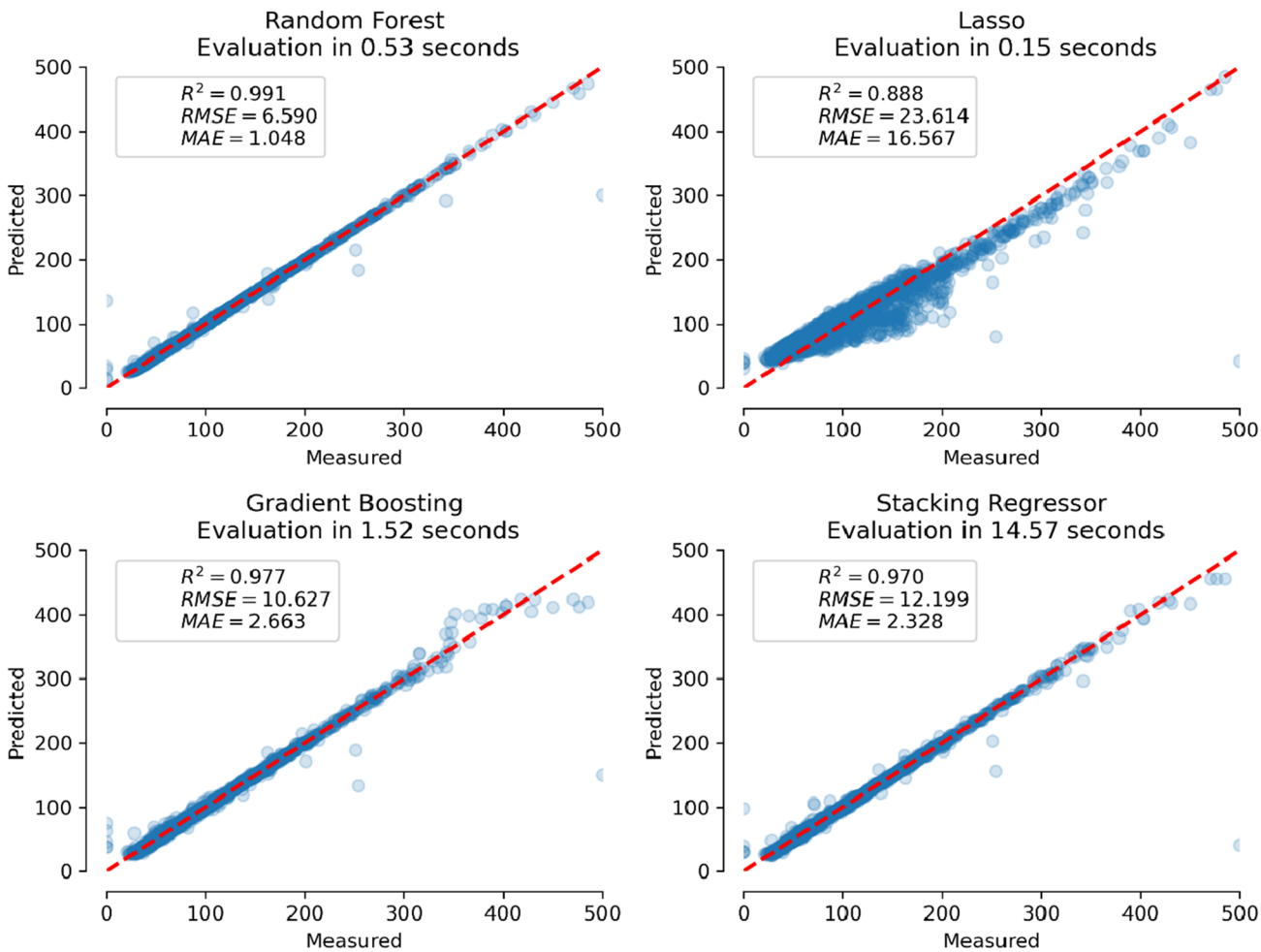

Fig. 6 The training result for the individual models and stack model

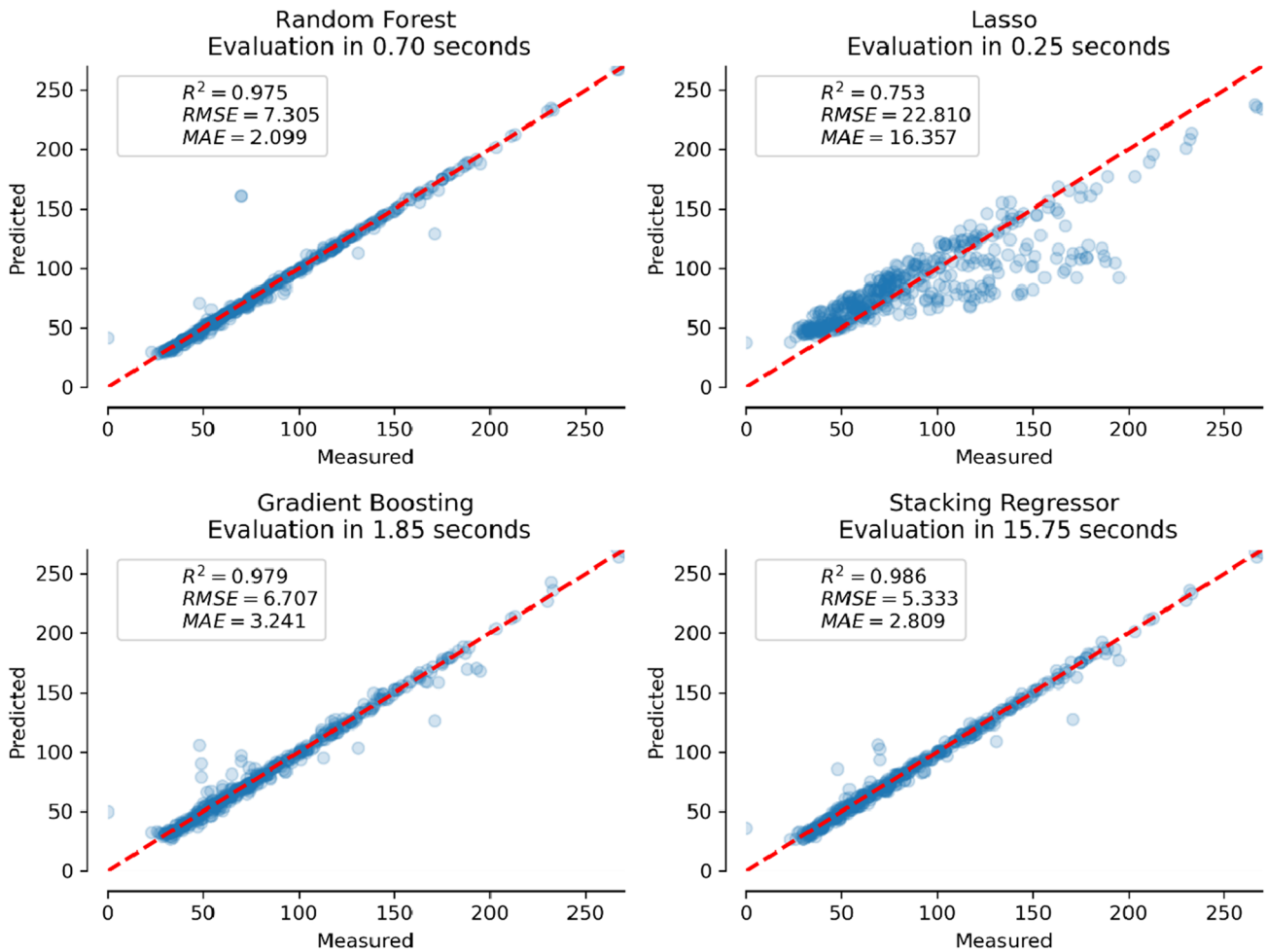

Fig. 7 Testing results for the individual models and the stack model

and precision for the training dataset in ascending order as LASSO < GB < RF.

Comparing the individual models to the stack model, which is an ensemble model created using a stacking regressor, the stack model was ranked third to RF and GB for the training set in terms of all performance evaluation metrics (R^2 and RMSE), except for MAE where the stack model had a lower error than GB as illustrated in Fig. 6.

In the testing datasets (see Fig. 7), GB achieved the highest score for explaining the proportion of variance in AQI explained by the independent variables ($R^2 = 0.979$). GB also had the least prediction error (RMSE = 6.707), while RF had the least magnitude of difference between AQI prediction and true values (MAE = 2.099). The models showed only marginal differences in performance. As shown in Table 2, RF had a 1.6% decrease in R^2, a 100% decrease in RMSE, and an 11% increase in MAE for training and testing, while LASSO had a 10%, 9%, and 1.6% decrease in R^2, RMSE, and MAE respectively. GB, however, had a 0.2% and 22% decrease in R^2 and MAE respectively, and a 37% increase in RMSE. Overall, the evaluation of individual models for the testing dataset showed that GB performed better

Table 3 Training and testing results for the individual models and the stack model

Models	Training set			Test set			Percent change		
	R^2	RMSE	MAE	R^2	RMSE	MAE	R^2%	RMSE %	MAE %
RF	0.991	6.590	1.048	0.975	7.305	2.099	1.6	11	100
LASSO	0.913	20.781	13.112	0.827	19.073	12.903	10	9	1.6
GB	0.977	10.627	2.663	0.979	6.707	3.241	0.2	37	22
Stack	0.968	12.667	2.471	0.981	6.253	3.040	1.3	102	23

than all other models across two evaluation scores, while LASSO had the worst performance across all metrics. The order of magnitude of accuracy and precision for the testing dataset in ascending order for individual models were LASSO < GB < RF.

In comparison to the stack model for the testing dataset, the stack model outperformed all individual models across different evaluation scores except for MAE, where RF had a lower score than the stack model. It is however noteworthy that there was an increase in the performance of the stack model from the training to the testing datasets as illustrated in Table 3.

These results indicate that although some individual models had better performance in each of the evaluation scores, they all had limitations. The stack model leveraged the strengths of the individual models to mitigate their weaknesses, demonstrating the generalizability and effectiveness of the ensemble model. Therefore, the stack model consistently performed better than the individual models across all evaluation metrics.

Figure 8 shows the distribution of errors in each of the models. In total, the stack model was the best model for forecasting AQI based on pollutant concentrations in this study. This finding is similar to that of Liang et al. (2020) who built a stacked ensemble model using KNN, SVM, DT, RF, and MLP for three cities in Taiwan (Zhongli, Changhua, Fengshan). They found that the stack model performed better than the individual models for all their cities and also for the training and the test set. Zhai and Chen (2018) also investigated the prediction of $PM_{2.5}$ with a stacked ensemble of Lasso, Adaboost, XGBoost, GA-MLP, and SVR. The following individual models obtained minimal error for the training set: XGBost (MAE, MAPE), Adaboost (RMSE), and SVM (Bias). However, the stacked ensemble obtained a minimal error for all error measurements with a high R^2 value compared with the individual models.

Comparison of AQG models

From Table 4, the RF model had the highest classification score (ACC = 0.998) among all the individual models in the training dataset. This indicates that the RF model was the most accurate in predicting the AQG and their corresponding actual values for the training set. Furthermore, the RF

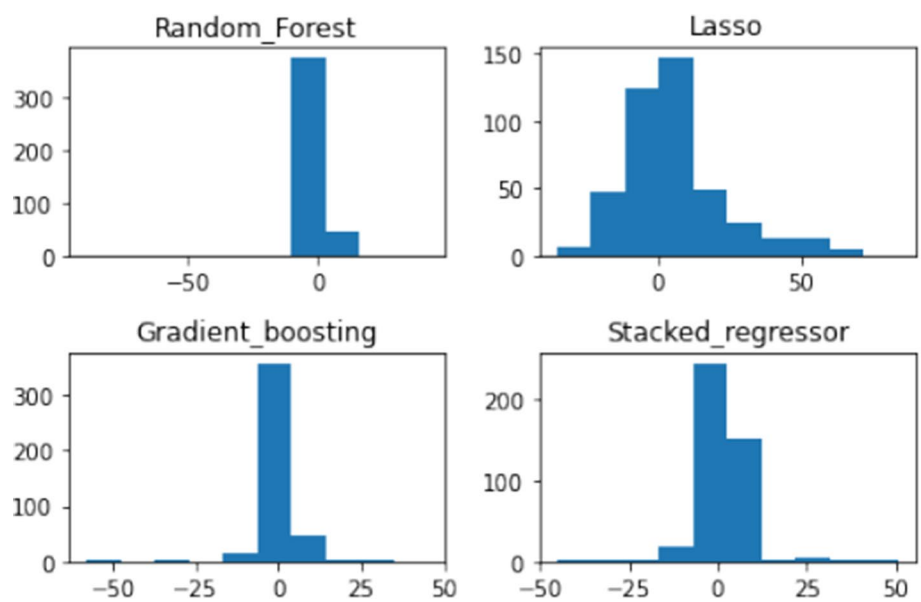

Fig. 8 The distribution of errors in each of the models

Table 4 Training and testing evaluation scores for AQG models

Models	Training set			Test set			Percent change		
	ACC	MCC	F1	ACC	MCC	F1	ACC %	MCC %	F1%
KNN	0.935	0.913	0.935	0.914	0.886	0.915	2.3	3.1	2.1
SVM	0.968	0.957	0.968	0.930	0.907	0.930	4	5	4
DT	0.979	0.972	0.979	0.963	0.951	0.963	1.6	2	1.6
RF	0.999	0.998	0.999	0.965	0.953	0.965	3.4	4.5	3.4
MLP	0.929	0.904	0.929	0.868	0.821	0.867	6.5	8.9	6.7
Stack	0.995	0.993	0.995	0.970	0.960	0.970	2.5	2.3	2.5

model had the highest balanced measure (MCC = 0.998) as well as precision and recall (F1 = 0.999). In contrast, the MLP model had the lowest scores for all the evaluation metrics used. It had an ACC of 0.929, an MCC of 0.904, and an F1 of 0.929. Overall, the evaluation of the individual models for the training set showed that the RF model performed better than all the other models across all evaluation scores. The accuracy and precision for the individual models in the training dataset in ascending order are MLP < KNN < SVM < DT < RF.

Comparing the performance of the individual models to the stack model (built using a logistic regression base classifier) for the training dataset, the stack model outperformed the individual models, except for RF (ACC = 0.999, MCC = 0.998, F1 = 0.999). The evaluation metric scores for the stack model in the training dataset were 0.995 for ACC, 0.993 for MCC, and 0.995 for F1.

For the testing datasets for the individual models (see Table 4), the RF model had the highest classification score (ACC = 0.953), balanced measure (MCC = 0.953), and precision and recall (F1 = 0.965), just as it did for the training set. However, all models showed a decrease in performance for the testing dataset compared to the training dataset. The percentage decrease in performance was 2.3%, 3.1%, and 2.1% for KNN; 4%, 5%, and 4% for SVM; 1.6%, 2%, and 1.6% for DT; 3.4%, 4.5%, and 3.4% for RF; and 6.5%, 8.9% and 6.7% for MLP, for ACC, MCC, and F1, respectively. The order of magnitude of accuracy and precision for the individual models in the testing dataset, in ascending order, are MLP < KNN < SVM < DT < RF. Overall, the RF model performed the best among all the individual models for both the training and testing datasets. This implies that RF was the best model for predicting the classes of AQG, compared to DT, MLP, SVM, and KNN.

In comparison to the stack model for the testing dataset (see Table 4), none of the individual models performed better than the stack model. The metric scores for the stack model were 0.970 for ACC, 0.960 for MCC, and 0.970 for F1. However, there was a reduction in the performance of the stack model from the training to the testing datasets. The evaluation scores for the testing model saw a 2.5%, 3.3%, and 2.5% reduction for ACC, MCC, and F1, respectively.

Overall, the findings from the performance of the individual models and the stack model showed that the stack model significantly improves the performance of predicting AQG compared to the single models. Therefore, the stack model was the best model for classifying AQG into categories of excellent, good, light pollution, moderate pollution, serious pollution, and severe pollution.

From this study, it is confirmed that the accuracy and efficiency of single machine learning models in forecasting AQI and AQG can be upgraded with stacked base regressors and classifiers. The stacked regressor and classifier can reduce bias and mitigate the concerns of overfitting against individual models. This enhances the performance of the models (Kuncheva 2014; Hong et al. 2018). The better performance of the stack model may be due to its ability to analyze how the errors evolve with the progression of the stack regressor or classifier which helps to detect overfitting (Prasad et al. 2021). Tables 5 and 6 show the actual and predicted AQI and AQG of sampled test results for all the models and Fig. 9 presents the residual plot of the actual and predicted AQI of the test dataset.

Table 5 Actual and predicted AQI of sampled test results for all models

	Actual AQI	RF	LASSO	GB	Stack model
0	34.0	33.92	47.15	33.78	31.60
1	30.0	30.64	45.32	31.17	26.80
2	43.60	36.57	49.20	33.50	37.20
3	48.0	47.87	65.64	48.32	48.40
4	54.0	53.50	76.83	55.25	54.20
…	…	…	…	…	…
426	183.0	182.65	167.15	184.66	183.40
427	175.0	174.98	159.60	175.02	175.20
428	30.0	30.12	47.50	31.83	30.80
429	40.0	35.93	49.00	38.52	35.60
430	73.0	70.87	73.09	72.39	67.40

Conclusion

This study used linear regression and logistic regression as a stacking regressor and classifier respectively with other machine learning models (KNN, SVM, DT, MLP, GB, and LASSO) for a single-step (one-day) AQI and AQG forecasting. Furthermore, the accuracy and prediction of the 6 individual models were compared with the stack model. The most important features for forecasting air quality index and air quality grade in order of importance are as follows: $PM_{2.5} > PM_{10} > O_3 > CO > NO_2 > SO_2$. For AQI forecasting, the stack model ($R^2 = 0.973$; RMSE = 7.568; MAE = 4.596) performed consistently across all metric scores as compared to LASSO, GB, and RF. This indicates that the stack model harnessed the strength of the individual models to minimize the effects of their weaknesses on forecasting. Similarly, for the AQG, the stack model (ACC = 0.970, MCC = 0.960, F1 = 0.970) performed better across all metric scores as compared to the MLP, KNN, SVM, DT, and RF. Hence the best model for classifying air quality grades (i.e., Excellent, good, light pollution, moderate pollution, serious pollution, and severe pollution). Stack generalisation machine learning models can be used to forecast AQI and AQG with high efficiency and precision as compared to individual

Table 6 Actual and predicted AQG of sampled test results for all models

	Actual AQG	KNN	SVM	DT	RF	MLP	Stack Model
0	4	4	4	4	4	4	4
1	5	5	5	5	5	5	5
2	4	4	4	4	4	4	4
3	3	3	3	3	3	3	3
4	3	3	3	3	3	3	3
…	…	…	…	…	…	…	…
426	5	5	5	5	5	5	5
427	4	4	4	4	4	4	4
428	0	0	0	0	0	0	0
429	4	4	4	4	4	4	4
430	3	2	2	3	3	3	3

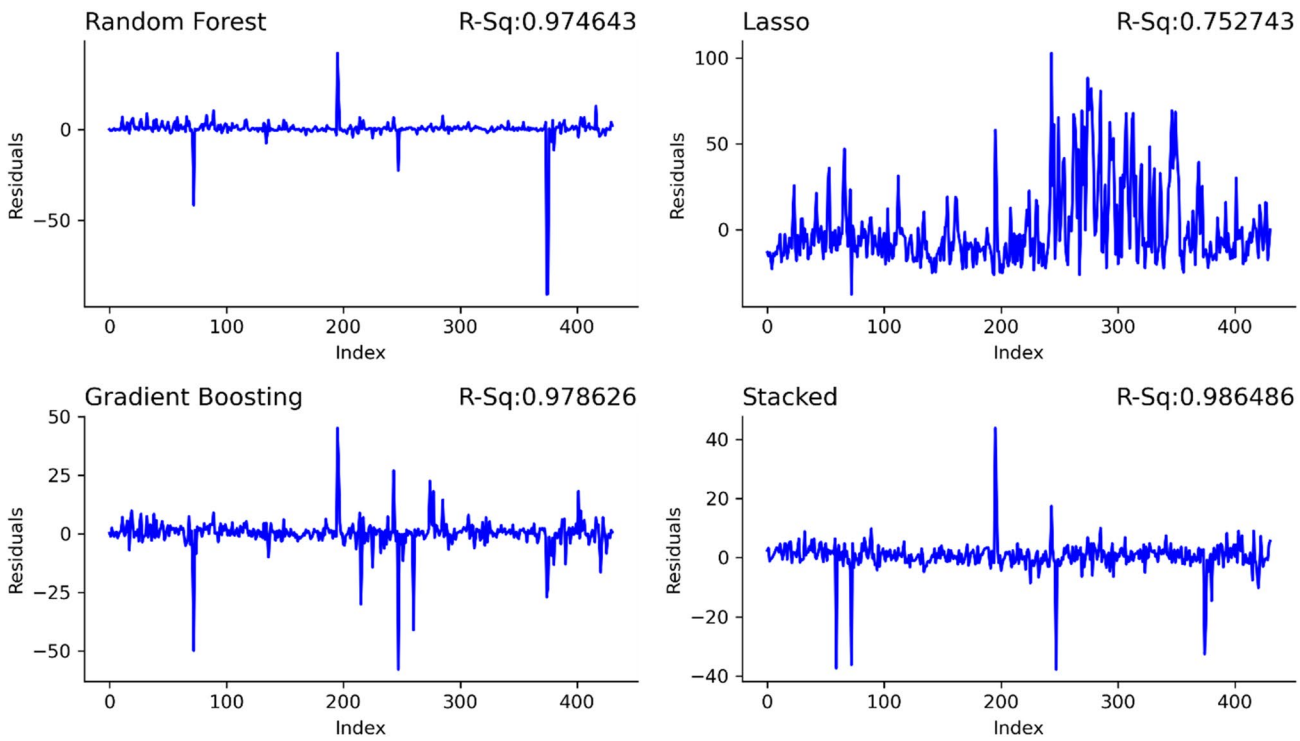

Fig. 9 Residual plot of the predicted AQI and test data

machine learning models. The stack regressor and classifier can reduce bias and mitigate the concerns of overfitting against individual models. In total, the stack model was the best model for forecasting one-day AQI and AQG based on current pollutant concentrations. Some developing countries still struggle with air quality forecasting, therefore, the proposed models in this study can be useful for air quality management with similar environmental factors across the world.

Funding The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Availability of data and materials The entire dataset used in this study was obtained from the China National Environmental Monitoring Centre (CNEMC, http://www.cnemc.cn).

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

References

Abelsohn A, Stieb D, Sanborn MD, Weir E (2002) Identifying and managing adverse environmental health effects: 2. Outdoor air pollution. CMAJ 166(9):1161–1167

Ahmadi K, Kalantar B, Saeidi V, Harandi EK, Janizadeh S, Ueda N (2020) Comparison of machine learning methods for mapping the stand characteristics of temperate forests using multi-spectral sentinel-2 data. Remote Sens 12(18):3019

Akinfolarin OM, Boisa N, Obunwo C (2017) Assessment of particulate matter-based air quality index in Port Harcourt Nigeria. J Environ Anal Chem 4(4):224

Alfeilat HAA, Hassanat AB, Lasassmeh O, Tarawneh AS, Alhasanat MB, Eyal Salman HS, Prasath VS (2019) Effects of distance measure choice on k-nearest neighbor classifier performance: a review. Big Data 7(4):221–248

Banerjee T, Srivastava RK (2011) Assessment of the ambient air quality at the Integrated Industrial Estate-Pantnagar through the air quality index (AQI) and exceedence factor (EF). Asia-Pac J Chem Eng 6(1):64–70

Bao J, Yang X, Zhao Z, Wang Z, Yu C, Li X (2015) The spatial-temporal characteristics of air pollution in China from 2001–2014. Int J Environ Res Public Health 12(12):15875–15887

Ben Seghier MEA, Carvalho H, Keshtegar B, Correia JA, Berto F (2020) Novel hybridized adaptive neuro-fuzzy inference system models based particle swarm optimization and genetic algorithms for accurate prediction of stress intensity factor. Fatigue Fract Eng Mater Struct 43(11):2653–2667

Breiman L (1996) Stacked regressions. Mach Learn 24(1):49–64

Bui DT, Tsangaratos P, Ngo PTT, Pham TD, Pham BT (2019) Flash flood susceptibility modeling using an optimized fuzzy rule based feature selection technique and tree based ensemble methods. Sci Total Environ 668:1038–1054

Canbek G, TaskayaTemizel T, Sagiroglu S (2021) BenchMetrics: a systematic benchmarking method for binary classification performance metrics. Neural Comput Appl 33(21):14623–14650

Chen B, Lu S, Li S, Wang B (2015) Impact of fine particulate fluctuation and other variables on Beijing's air quality index. Environ Sci Pollut Res 22(7):5139–5151

Cheng Y, Zhang H, Liu Z, Chen L, Wang P (2019) Hybrid algorithm for short-term forecasting of PM2. 5 in China. Atmos Environ 200:264–279

Chicco D, Jurman G (2020) The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. BMC Genom 21(1):1–13

Choi G, Heo S, Lee J-T (2016) Assessment of environmental injustice in Korea using synthetic air quality index and multiple indicators of socioeconomic status: a cross-sectional study. J Air Waste Manag Assoc 66(1):28–37

Dastoorpoor M, Idani E, Goudarzi G, Khanjani N (2018) Acute effects of air pollution on spontaneous abortion, premature delivery, and stillbirth in Ahvaz, Iran: a time-series study. Environ Sci Pollut Res 25(6):5447–5458

Dominici F, Peng RD, Barr CD, Bell ML (2010) Protecting human health from air pollution: shifting from a single-pollutant to a multi-pollutant approach. Epidemiology 21(2):187–194

Dragomir EG (2010) Air quality index prediction using K-nearest neighbor technique. Bulletin of PG University of Ploiesti, Series Mathematics, Informatics, Physics, LXII 1(2010):103–108

Feng Q, Wu S, Du Y, Xue H, Xiao F, Ban X, Li X (2013) Improving neural network prediction accuracy for PM10 individual air quality index pollution levels. Environ Eng Sci 30(12):725–732

Ganesh SS, Modali SH, Palreddy SR, Arulmozhivarman P (2017) Forecasting air quality index using regression models: a case study on Delhi and Houston. 248–254

GB 3095–2012 (2012) China Ambient air quality standards. Environmental Development Center

Harrington P (2012) Machine learning in action. Simon and Schuster

Ho TK (1995) Random decision forests. In: Proceedings of 3rd international conference on document analysis and recognition, vol 1, pp 278–282

Hong H, Liu J, Bui DT, Pradhan B, Acharya TD, Pham BT, Ahmad BB (2018) Landslide susceptibility mapping using J48 decision tree with AdaBoost, bagging and rotation forest ensembles in the Guangchang area (China). CATENA 163:399–413

Janarthanan R, Partheeban P, Somasundaram K, Elamparithi PN (2021) A deep learning approach for prediction of air quality index in a metropolitan city. Sustain Cities Soc 67:102720

Kamiński B, Jakubczyk M, Szufel P (2018) A framework for sensitivity analysis of decision trees. Cent Eur J Oper Res 26(1):135–159

Khosravi K, Pham BT, Chapi K, Shirzadi A, Shahabi H, Revhaug I, Bui DT (2018) A comparative assessment of decision trees algorithms for flash flood susceptibility modeling at Haraz watershed, northern Iran. Sci Total Environ 627:744–755

Kumar A, Goyal P (2011a) Forecasting of daily air quality index in Delhi. Sci Total Environ 409(24):5517–5523. https://doi.org/10.1016/j.scitotenv.2011.08.069

Kumar A, Goyal P (2011b) Forecasting of air quality in Delhi using principal component regression technique. Atmos Pollut Res 2(4):436–444

Kuncheva LI (2014) Combining pattern classifiers: methods and algorithms. John Wiley and Sons, New Jersey

Kurt A, Oktay AB (2010) Forecasting air pollutant indicator levels with geographic models 3 days in advance using neural networks. Expert Syst Appl 37(12):7986–7992

Liang YC, Maimury Y, Chen AHL, Juarez JRC (2020) Machine learning-based prediction of air quality. Appl Sci 10(24):9151

Liu H, Chen C (2020) Spatial air quality index prediction model based on decomposition, adaptive boosting, and three-stage feature selection: a case study in China. J Clean Prod 265:121777

Liu H, Li Q, Yu D, Gu Y (2019) Air quality index and air pollutant concentration prediction based on machine learning algorithms. Appl Sci 9(19):4069

Mahalingam U, Elangovan K, Dobhal H, Valliappa C, Shrestha S, Kedam G (2019) A machine learning model for air quality prediction for smart cities. In: International conference on wireless communications signal processing and networking (WiSPNET) pp 452–457. IEEE

Maynard R (2004) Key airborne pollutants—the impact on health. Sci Total Environ 334:9–13

Mishra D, Goyal P (2015) Analysis of ambient air quality using fuzzy air quality index: a case study of Delhi, India. Int J Environ Pollut 58(3):149–159

Mohammed Y, Caleb J (2014) Assessment of some air pollutants and their corresponding air quality at selected activity areas in Kaduna metropolis. In: Paper presented at the Proceedings of 37th Annual International Conference of Chemical of Nigeria (SCN) at Uyo, Akwa Ibom State Nigeria, 7th

Pan R, Wang X, Yi W, Wei Q, Gao J, Xu Z, Duan J, He Y, Tang C, Liu X, Zhou Y, Son S, Ji Y, Zou Y, Su H (2020) Interactions between climate factors and air quality index for improved childhood asthma self-management. Sci Total Environ 723:137804. https://doi.org/10.1016/j.scitotenv.2020.137804

Polley E, LeDell E, Kennedy C, van der Laan MS (2019) Super learner prediction. 2018. URL http://CRAN.R-project.org/package=SuperLearner, r package version, 2–0

Prasad P, Loveson VJ, Das B, Kotha M (2021) Novel ensemble machine learning models in flood susceptibility mapping. Geocarto Int 1–23

Qiao X, Jaffe D, Tang Y, Bresnahan M, Song J (2015) Evaluation of air quality in Chengdu, Sichuan Basin, China: are China's air quality standards sufficient yet? Environ Monit Assess 187(5):1–11

Sicard P, Lesne O, Alexandre N, Mangin A, Collomp R (2011) Air quality trends and potential health effects–development of an aggregate risk index. Atmos Environ 45(5):1145–1153

Soni, HB, Patel J (2018) Assessment of Ambient Air Quality and Air Quality Index in Golden Corridor of Gujarat, India: a case study of Dahej port. Int J Environ

Sonibare J, Adebiyi F, Obanijesu E, Okelana O (2010) Air quality index pattern around petroleum production facilities. Manag Environ Qual Int J

Sowlat MH, Gharibi H, Yunesian M, Mahmoudi MT, Lotfi S (2011) A novel, fuzzy-based air quality index (FAQI) for air quality assessment. Atmos Environ 45(12):2050–2059

Taylan O (2017) Modeling and analysis of ozone concentration by artificial intelligent techniques for estimating air quality. Atmos Environ 150:356–365

Wang H, Jiao M, Tan Y (2016) Air quality index forecast based on fuzzy time series models. J Resid Sci Technol 13(5)

Wolpert DH (1992) Stacked generalization. Neural Netw 5(2):241–259

World Health Organization (2013) Health Effects of particulate matter: policy implications for countries in eastern Europe, Caucasus and central Asia

Wu Q, Lin H (2019) A novel optimal-hybrid model for daily air quality index prediction considering air pollutant factors. Sci Total Environ 683:808–821

Zhai B, Chen J (2018) Development of a stacked ensemble model for forecasting and analyzing daily average PM2.5 concentrations in Beijing China. Sci Total Environ 635:644–658

Zhang X, Lu JJ, Qin X, Zhao XN (2013) A high-level energy consumption model for heterogeneous data centers. Simul Model Pract Theory 39:41–55

Zhu S, Lian X, Liu H, Hu J, Wang Y, Che J (2017) Daily air quality index forecasting with hybrid models: a case in China. Environ Pollut 231(Pt 2):1232–1244. https://doi.org/10.1016/j.envpol.2017.08.069

Zhu S, Qiu X, Yin Y, Fang M, Liu X, Zhao X, Shi Y (2019) Two-step-hybrid model based on data preprocessing and intelligent optimization algorithms (CS and GWO) for NO2 and SO2 forecasting. Atmos Pollut Res 10(4):1326–1335

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